Stat 266 – Unit 8 – Video Set 2 – Class Examples

Summary:

Definition of Consistency :

The estimator $\hat{\theta}_n$ is said to be a **consistent estimator** for θ if, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} P\left(\left| \underbrace{\hat{\theta}_n - \theta}_{\text{error in estimation}} \right| \le \varepsilon\right) = 1.$$

Comment : If $\hat{\theta}_n$ is a consistent estimator for θ then we say that $\hat{\theta}_n$ converges in probability to θ, which I will denote as $\hat{\theta}_n \xrightarrow{P} \theta$.

Theorem: The unbiased estimator $\hat{\theta}_n$ is a **consistent estimator** for θ if $\lim_{n\to\infty} V(\hat{\theta}_n) = 0$.

Does the estimator improve as we increase n?

Example A:

Suppose $Y_1, \ldots, Y_n$ is a random sample of independent and identically distributed random variables with density function given by

Parameter. Assume α is a known constant.

$$f_Y(y) = \begin{cases} \dfrac{\alpha\beta^\alpha}{y^{\alpha+1}}, & 0 < \beta < y, \alpha > 0 \\ 0 & \text{, else} \end{cases}$$

β = population minimum

Find $P\left(\left|Y_{(1)} - \beta\right| \le c\right)$, where $c > 0$ is a positive constant. Based on this probablity, is $Y_{(1)}$ a consistent estimator for β?

$$f_{Y_{(1)}}(y) = n\left[1 - F_Y(y)\right]^{n-1} f_Y(y)$$

$$\left(F_Y(y) = P(Y \le y) = \int_\beta^y \frac{\alpha\beta^\alpha}{t^{\alpha+1}}\,dt = 1 - \left(\frac{\beta}{y}\right)^\alpha, \quad y \ge \beta \right)$$

$$= n\left[\left(\frac{\beta}{y}\right)^\alpha\right]^{n-1} \cdot \frac{\alpha\beta^\alpha}{y^{\alpha+1}} = \frac{\alpha n \beta^{\alpha n}}{y^{\alpha n + 1}}, \quad y > \beta.$$

Also, $$F_{Y_{(1)}}(y) = \int_\beta^y \frac{\alpha n \beta^{\alpha n}}{t^{\alpha n + 1}}\,dt = 1 - \left(\frac{\beta}{y}\right)^{\alpha n}, \quad y \ge \beta.$$

$$P(|Y_{(1)} - \beta| \le c) = P(-c \le Y_{(1)} - \beta \le c)$$

$$= P(\underbrace{\beta - c}_{\substack{\beta - c < \beta \\ \text{since } c > 0}} \le Y_{(1)} \le \beta + c) = P(\beta \le Y_{(1)} \le \beta + c)$$

$$= \int_{\beta}^{\beta+c} \frac{\alpha n \beta^{\alpha n}}{y^{\alpha n + 1}} \, dy \quad \text{or} \quad F_{Y_{(1)}}(\beta + c) - F_{Y_{(1)}}(\beta)$$

$$= -\beta^{\alpha n} y^{-\alpha n} \Big]_{\beta}^{\beta+c} = 1 - \left(\frac{\beta}{\beta + c}\right)^{\alpha n}.$$

$$\lim_{n \to \infty} \left(1 - \underbrace{\left(\frac{\beta}{\beta + c}\right)}_{0 < \frac{\beta}{\beta + c} < 1}^{\alpha n}\right) = 1.$$

So $Y_{(1)}$ is consistent for β.

- $E[Y_{(1)}] = \int_{\beta}^{\infty} y\, f_{Y_{(1)}}(y)\, dy = \int_{\beta}^{\infty} \frac{\alpha n \beta^{\alpha n}}{y^{\alpha n}}\, dy = \frac{\alpha n}{\alpha n - 1}\beta \neq \beta$

So $\frac{\alpha n - 1}{\alpha n} Y_{(1)}$ is unbiased for β.

- $E[Y_{(1)}^2] = \int_{\beta}^{\infty} \frac{\alpha n \beta^{\alpha n}}{y^{\alpha n - 1}}\, dy = \frac{\alpha n}{\alpha n + 2}\beta^2.$

- $V\left[\frac{\alpha n - 1}{\alpha n} Y_{(1)}\right] = \left(\frac{\alpha n - 1}{\alpha n}\right)^2 \left(\underbrace{\frac{\alpha n}{\alpha n + 2}\beta^2 - \left(\frac{\alpha n}{\alpha n - 1}\beta\right)^2}_{V[Y_{(1)}] = E[Y_{(1)}^2] - (E[Y_{(1)}])^2}\right)$

$$= \frac{1}{\alpha n(\alpha n - 2)}\beta^2.$$

$$\lim_{n \to \infty} \frac{1}{\alpha n(\alpha n - 2)}\beta^2 = 0.$$

$\frac{\alpha n - 1}{\alpha n} Y_{(1)}$ is unbiased and consistent for β.

Example B:

Suppose $X_1, \ldots, X_n$ is a random sample of independent and identically distributed $Uniform(0, \theta)$ random variables.

Consider the unbiased estimators

$$\hat{\theta}_1 = \left(\frac{n+1}{n}\right) X_{(n)}, \quad \hat{\theta}_2 = (n+1) X_{(1)}, \text{ and } \hat{\theta}_3 = 2\bar{X} \quad \text{for } \theta.$$

Which of the estimators are consistent for θ?

Parameter: θ = population maximum.

$$V[\hat{\theta}_1] = \frac{1}{n(n+2)} \theta^2 \to 0 \text{ as } n \to \infty \quad \text{(consistent)}$$

$$V[\hat{\theta}_2] = \frac{n}{n+2} \theta^2 \to \theta^2 \text{ as } n \to \infty \quad \text{(not consistent)}$$

$$V[\hat{\theta}_3] = \frac{1}{3n} \theta^2 \to 0 \text{ as } n \to \infty \quad \text{(consistent)}$$

<u>Note</u>: $\mu_x = \theta/2$

$\frac{n+1}{2n} X_{(n)}$ is a better estimator for μ_x than $\bar{X}$.